

A real scalar completion when all but one columns coincide

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Lemma 1 (An ordered root-step criterion). *Let $m \geq 2$, $0 \leq a_1 < \dots < a_m \leq 1$, and $P(z) = \prod_{i=1}^m (z - a_i)$. If for a real δ the numbers*

$$b_i = a_i + \frac{m\delta}{P'(a_i)}$$

satisfy $0 \leq b_1 \leq \dots \leq b_m \leq 1$, then $P - \delta$ has all its m roots in $[0, 1]$, counted with multiplicity.

Proof. The case $\delta = 0$ is immediate. Otherwise put $\sigma = \text{sgn } \delta$, $R = \sigma P$, and $\epsilon = |\delta|$. Then $b_i = a_i + m\epsilon/R'(a_i)$.

Consider an interval $(u, v) = (a_i, a_{i+1})$ where R is positive. Set $h = v - u$, $A = R'(u) > 0$, and $B = -R'(v) > 0$. The inequality $b_i \leq b_{i+1}$ gives

$$\epsilon \leq \frac{hAB}{m(A+B)}.$$

On this interval write $R(z) = (z - u)(v - z)T(z)$, where $T > 0$ is a product of absolute distances to the other real roots. Its logarithm is concave, so

$$R\left(\frac{u+v}{2}\right) = \frac{h^2}{4}T\left(\frac{u+v}{2}\right) \geq \frac{h\sqrt{AB}}{4} \geq \frac{hAB}{2(A+B)} \geq \epsilon.$$

Thus $R - \epsilon$ has two zeros in $[u, v]$, counted with multiplicity.

If R is positive to the left of a_1 , put $A = -R'(a_1) > 0$. From $b_1 \geq 0$ we get $\epsilon \leq a_1 A/m$, while

$$R(0) = \prod_i a_i \geq a_1 \prod_{i>1} (a_i - a_1) = a_1 A \geq \epsilon.$$

Hence $R - \epsilon$ has a zero in $[0, a_1]$. The same argument at the right endpoint uses $b_m \leq 1$ and gives a zero in $[a_m, 1]$ whenever R is positive to the right of a_m . Every root a_i bounds exactly one positive component of R . The resulting count is therefore m zeros of $R - \epsilon$, all in $[0, 1]$. Its zeros are those of $P - \delta$. $\square$

Theorem 2. *Let $p, q : [m] \rightarrow [0, 1]$ be nondecreasing, with uniform averaging on $[m]$. Suppose*

$$\mathbb{E}p^a = \frac{1}{a+1} \quad (0 \leq a \leq m-1), \quad \mathbb{E}(p^a q) = \frac{1}{a+2} \quad (0 \leq a \leq m-1).$$

Then there exists a real equal-weight uniform-interval quadrature of degree m with m nodes in $[0, 1]$. In particular this applies to a monotone comparison tensor on $K = m$ atoms when at least $m - 1$ of its m columns coincide.

Proof. Put $e = q - p$ and $\delta = 1/(m+1) - \mathbb{E}p^m$. The assumptions give

$$\mathbb{E}(p^a e) = 0 \quad (0 \leq a \leq m-2), \quad \mathbb{E}(p^{m-1} e) = \delta.$$

If p has fewer than m distinct values, interpolate on these values to see that p^{m-1} lies in the span of $1, p, \dots, p^{m-2}$. Then $\delta = 0$, so the values of p themselves give the rule.

Otherwise write $a_i = p(i)$ and $P(z) = \prod_i (z - a_i)$. The Vandermonde identities

$$\sum_i \frac{a_i^j}{P'(a_i)} = \begin{cases} 0, & 0 \leq j \leq m-2, \\ 1, & j = m-1 \end{cases}$$

show uniquely that $q(i) = a_i + m\delta/P'(a_i)$. Lemma 1 says that $Q = P - \delta$ has all its roots in $[0, 1]$. Newton's identities show that changing only the constant coefficient from that of P to that of $P - \delta$ leaves the first $m-1$ power sums unchanged and adds $m\delta$ to the m th power sum. The m roots of Q therefore have precisely the required moments through degree m . $\square$

Scope. The replacement nodes need not be rational, even if p and q are. The theorem only treats $K = m$ with $m-1$ coincident columns; it does not establish a scalar reduction for arbitrary monotone comparison tensors. Such a general existence reduction would imply a quadratic lower bound through the classical lower bound for real equal-weight uniform-interval quadrature. No claim of that general reduction, or of literature novelty for the root-step criterion, is made here.